RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter S2B: Binomial Distribution

1 Published articles in medical journals indicate that, on average, 35 out of 100 patients having a lumbar puncture will suffer SSH ('Severe Spinal Headache'). Twelve patients are given a lumbar puncture.

(i) Using a binomial model, find the expected number of patients who will suffer SSH, and find also the standard deviation. [3]

(ii) Find the probability that four or more of the twelve patients will suffer SSH. [2]

[**(i)** 4.2, 1.65 **(ii)** 0.653]

2 The random variable $X \sim \text{B}(16, p)$, where $p < 0.5$. If the variance of X is 3.36, find the value of p. Find also the probability that X is less than its mean.

[$p = 0.3$, 0.450]

3 A factory produces chocolate which are packed into boxes of 20 and delivered to shops for sale. A chocolate will not meet the minimum criteria for packing for sale if it weighs less than 20 grams. On average, 2% of the chocolate produced did not meet the minimum criteria.

(i) Find the probability that a randomly chosen box contain at least 1 chocolate that does not meet the minimum criteria. [2]

(ii) Find the probability that out of 4 randomly chosen boxes of chocolate, there are exactly 2 boxes with at least 1 chocolate that does not meet the minimum criteria. [2]

[**(i)** 0.332 **(ii)** 0.295]

4 In a multi-national company with a large population, 13.5% of the staff owns a vehicle.

(i) Find the probability that in a random sample of 30 staff, exactly 5 of them will own a vehicle. [2]

(ii) The probability that there is at least one staff who owns a vehicle in a random sample of size n is greater than 0.95. Find the least value of n. [3]

[**(i)** 0.170 **(ii)** 21]

5 Market research showed that 3 out of 10 households in a housing estate subscribe to fibre broadband internet services.

(a) 20 households from a particular block of flats in the estate were surveyed.

(i) Show that the probability of more than 3 and less than 9 of the households surveyed subscribe to fibre broadband internet services is 0.780, correct to 3 decimal places. [2]

(ii) Find the least value of k such that the probability that at most k of the households surveyed subscribe to fibre broadband internet services is at least 0.75. [2]

(b) There are a total of 50 blocks of flats in the estate. 20 households from each block of flats were surveyed.
Find the probability that there are exactly 39 blocks of flats with more than 3 and less than 9 households surveyed that subscribe to fibre broadband services. [2]

[**(a)(ii)** least $k = 7$ **(b)** 0.135]

6 It is known that 36% of the customers of a certain supermarket will bring their own environmentally friendly bags. On a certain day, there are 3 cashiers and each cashier has 5 customers in queue.

(i) Find the probability that among all the customers in queue, at least 4 of them brought their own environmentally friendly bags. [2]

(ii) If exactly 4 customers in queue brought their own environmentally friendly bags, find the probability that each cashier will have at least 1 customer bringing his or her own environmentally friendly bag. [4]

[**(i)** 0.847 **(ii)** 0.549]

7 A factory produces watches. The probability that a watch is defective is 0.02. The watches are packed into boxes which contain 55 watches each before being shipped.

(i) State, in context, two assumptions for the number of defective watches in a box to be well modelled by a binomial distribution. [2]

(ii) Show that the probability for a box to contain at least 1 defective watch is 0.671. [2]

(iii) A customer ordered 40 boxes of watches. He will be compensated if more than 19 boxes contain defective watches. Find the probability that the customer will be compensated. [3]

(iv) Find the least number of watches to be added to a box of 55 watches such that the probability of finding at least 53 non-defective watches is more than 0.99. [3]

[**(iii)** 0.992 **(iv)** least number is 2]

8 A test consists of 15 multiple choice questions, where each question has n possible options, of which only one is correct. A student took the test by randomly choosing an answer to each question. It is known that the probability of answering exactly 3 questions correctly is the same as the probability of answering exactly 4 questions correctly.

(i) By forming an equation in terms of n, find the value of n. [3]

Each correct answer is awarded 3 marks and each incorrect answer carries a penalty of 1 mark. The score is the total marks awarded based on the number of correct and incorrect answers.

(ii) Find the expected score, s, obtained by the student. [3]

(iii) Find the probability that the score obtained by the student is within 4 marks of s. [2]

[**(i)** 4 **(ii)** 0 **(iii)** 0.450]

9 The random variable X is the number of successes in n independent trials of an experiment in which the probability of a success in any one trial is p. Show that

$$\frac{\mathrm{P}(X=k+1)}{\mathrm{P}(X=k)}=\frac{(n-k)p}{(k+1)(1-p)}, \quad k=0,1,2,...,(n-1)$$

Find the most probable number of successes when $n=10$ and $p=\frac{1}{4}$.

[most probable number = 2]

10 A game is played using a fair six-sided die, a pawn and a simple board as shown below.

S	1	2	3	4	5	**E**

Initially, the pawn is placed on square **S**. The game is played by throwing the die and moving the pawn in the following manner:

S 1 2 3 4 5 **E** 5 4 3 2 1 2 3 4 5 **E**……

Thus, for example, if the first and second throw of the die gives a "5" and "4" respectively, the final position of the pawn will be on square "3".

The game will stop when the pawn stops at square **E**.

Let X be the random variable denoting the number of throws of the die required to move the pawn such that it stops at square **E**.

(i) Show $\mathrm{P}(X=2)=\frac{5}{36}$. [1]

(ii) Find the probability that more than two throws of the die are needed for the pawn to stop at square **E** given that the first throw of the die gives an even number. [3]

It is now given that for each game, a player has a maximum of 3 throws of the die and a special prize is given to any player who uses not more than two throws for the pawn to stop at square **E**.

(iii) Find the probability of a player winning a special prize in at least three but not more than eight games out of ten games. [3]

(iv) Find the least number of games needed so that the probability of winning at least a special prize is at least 0.998.

[**(ii)** $\frac{5}{9}$ **(iii)** 0.632 **((iv)** 18)]

11 A factory produces ballpoint pens. On average 6% of the pens are faulty. The pens are packed in boxes of 100 for sale to retail outlets. It should be assumed that the number of faulty pens in a box of 100 pens follows a binomial distribution.

For quality control purposes a random sample of 10 pens from each box is tested. If 2 or fewer faulty pens are found in the sample of 10, the box is accepted for sale. Otherwise the box is rejected.

(i) Find the probability that a randomly chosen box of 100 pens is accepted for sale. [1]

(ii) One morning 75 boxes are tested in this way. Find the probability that more than 5% of these boxes are rejected. [4]

An alternative testing procedure is trialled in which a random sample of 5 pens is initially taken from a box and tested.

- If there are no faulty pens in this sample of 5, the box is accepted.
- If there are 3 or more faulty pens in this sample of 5, the box is rejected.
- If there are 1 or 2 faulty pens in this sample, a second random sample of 5 pens is taken from the box. When the second sample has been tested, the box is accepted if the total number of faulty pens found in the combined sample of 10 is 2 or fewer and rejected otherwise.

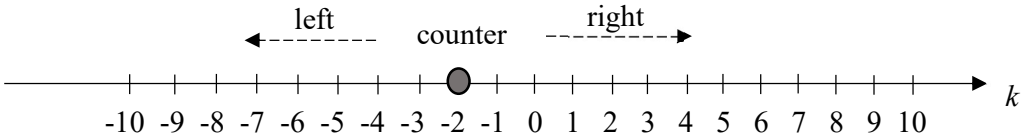
(iii) Find the probability that a randomly chosen box of 100 pens is accepted for sale when the alternative testing procedure is used. [5]

(iv) Explain why the factory manager might prefer to use the alternative testing procedure. [1]

[**(i)** 0.981 **(ii)** 0.0535 **(iii)** 0.983]

12 In a computer game, a counter moves along a straight line and is originally placed at $k = 0$. At each stage, it takes one step to the right with probability p or one step to the left with probability q, where $q = 1 - p$. Each step is of length 1 unit and the step taken at each stage is independent of one another.

For illustration purpose, the counter is seen to be at $k = -2$ in the diagram below.



The counter takes 10 consecutive steps.

(i) Explain why the counter cannot possibly end at $k = 7$. [1]

(ii) Show that the probability that the counter ends at $k = 6$ is $45p^8q^2$. [2]

(iii) Given that the most probable end-point of the counter is $k = 6$, find exactly the range of values of p in the form $p_1 < p < p_2$ where p_1 and p_2 are constants to be determined. [4]

(iv) Write down the most probable end-point(s) of the counter if $p = 1 - p_1$. [1]

[**(iii)** $p_1 = \frac{8}{11}, p_2 = \frac{9}{11}$ **(iv)** $k = -4, -6$]

THE END